Comenius University in Bratislava
Faculty of Mathematics, Physics and Informatics

Analysis of financial markets using machine learning

Bachelor Thesis

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Analysis of financial markets using machine learning

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Field of Study:	Computer Science and Mathematics
Department:	Department of Applied Mathematics and Statistics
Supervisor:	Mgr. Samuel Rosa, PhD.

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THESIS ASSIGNMENT

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Annotation: Modeling and prediction of stocks in financial markets have long been a researched problem. Historically, this problem was approached using specific methods of financial mathematics, but with the significant development and success of machine learning in general, machine learning methods have also begun to be applied to predicting the behavior of financial markets. The aim of this thesis is to create a prediction model for selected stocks, which involves the appropriate selection and identification of relevant explanatory variables, as well as the suitable choice of model. Subsequently, it will be possible to analyze whether the obtained predictions can be used for successful creation of an investment portfolio.

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Declaration: I hereby declare that I have independently written this entire bachelor thesis entitled "Analysis of financial markets using machine learning", including all its figures and appendices, using the literature listed in the attached bibliography.

In preparation of the thesis I have also used the LLM ChatGPT for the keyword-based searches in particular Python libraries' documentations. I have used artificial intelligence tools in accordance with relevant legal regulations, academic rights and freedoms, ethical and moral principles while simultaneously maintaining academic integrity. I am aware that I am fully responsible for the accuracy of the final text.

Acknowledgments: This way I would like to thank my supervisor Dr. Rosa, for his mentorship, spent time and intriguing feedback throughout the whole process of thesis' creation. Last but not least, I would like to express thanks to all my family for their endless support and motivation.

Abstrakt

Na tomto mieste sa bude nachádzať slovenský abstrakt

Kľúčové slová: strojové učenie, akciové trhy, prediktívna analýza

Abstract

Here will be English written abstract.

Keywords: machine learning, stock markets, predictive analysis

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Introduction

Here will be the Introduction chapter, where reader will be introduced to the topic of Quantitative Science, its novelty and to the topics being dealt with in academic community and practice as well.

Chapter 1

Machine learning foundations

The aim of this chapter is to introduce the methods of machine learning, which are going to be used for the purposes of predictive regression modeling in this thesis. The chapter starts with a brief overview of machine learning as a method, followed by descriptions of the individual algorithms used.

1.1 Machine learning as a method

Nowadays, we can say that machine learning (ML) has become a buzzworthy term in the field of data science. It stands for the subset of artificial intelligence enabled to identify patterns in data [2]. Generally speaking, ML implements algorithms capable of learning from data and generalizing patterns to be able to make predictions on unseen data, which means being able to perform tasks without being explicitly told to [8].

Mathematically speaking, ML builds on the idea of having a quantitative target variable Y and p various predictors $X_1, X_2, \ldots, X_p$. We assume the existence of an underlying function f such that

$$Y = f(X_1, X_2, \ldots, X_p) + \epsilon, \tag{1.1}$$

where ϵ is a random noise independent of $X_1, X_2, \ldots, X_p$. Algorithms and methods of ML aim to estimate the function f by using a so-called **training dataset** of observed data.

Suppose we have n observations of the introduced predictors and the corresponding target Y as $\{Y_i, X_{i,1}, X_{i,2}, \ldots, X_{i,p}\}_{i=1}^{n}$, where $X_{i,q}$ denotes value of the q-th predictor in the i-th observation. Then, ML algorithms use this dataset to approximate function $\hat{f}$, such that $\hat{f}(X_{i,1}, X_{i,2}, \ldots, X_{i,p}) = \hat{Y}_i \approx Y_i$, for all $i = 1, 2, \ldots, n$. With the obtained $\hat{f}$, we can make predictions of Y for new (unseen before) values of predictors $X_1, X_2, \ldots, X_p$.

Based on the methodology used to find $\hat{f}$, we can distinguish two methods – *parametric* and *non-parametric*. **Parametric methods** seek to approximate f by specifically optimizing a various number of parameters, which then explicitly define the function $\hat{f}$. This involves a prior assumption about the functional form of f (e.g., linear, polynomial, etc.). Then, using specific algorithms, the parameters of the function are estimated from the training data, resulting in $\hat{f}$. Examples of such methods used in this thesis are linear regression or its regularized variants.

On the other hand, **non-parametric methods** do not assume any specific functional form of f. Instead, they rely on patterns included in the data itself to roughly approximate f. Generalisation to unseen data is achieved by identifying patterns and similarities in the training data, without explicitly defining $\hat{f}$. An example of such a method used in this thesis is the k-nearest neighbours algorithm or the random forest algorithm.

Next, based on the character of the Y, we can distinguish *regression* and *classification* tasks. In **regression** tasks, Y is a continuous variable, while in **classification** tasks, Y is a discrete variable.

Lastly, in order to have a complete picture of ML theory foundations, it is important to mention the terms *overfit* and *underfit* learning. In the process of building the model to approximate f, we might encounter such a phenomenon, where the trained model captures patterns in the training dataset very well (memorizes), but, on the other hand, on a set of unseen data fails in terms of accuracy – this is called **overfit**. In contrast, **underfit** comes when the model and resulting $\hat{f}$ fail to generalize patterns on both training and unseen sets.

1.2 Linear regression

Linear regression is undoubtedly the most fundamental method in ML. It is a parametric method used for regression tasks, assuming that the true relationship between the predictors and the target variable is linear.

Given a training dataset of n observations $\{Y_i, X_{i,1}, X_{i,2}, \ldots, X_{i,p}\}_{i=1}^{n}$, where $X_{i,q}$ denotes the value of the q-th predictor in the i-th observation, we can formulate the linear relationship as

$$Y_i = \beta_0 + \beta_1 X_{i,1} + \ldots + \beta_p X_{i,p} + \epsilon_i, \qquad i = 1, 2, \ldots, n \tag{1.2}$$

with ϵ_i representing an independent noise. Generalising these equations to a more convenient matrix notation, we obtain

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \tag{1.3}$$

where

$$\mathbf{Y} = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix} \qquad \mathbf{X} = \begin{bmatrix} 1 & X_{1,1} & \dots & X_{1,p} \\ 1 & X_{2,1} & \dots & X_{2,p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n,1} & \dots & X_{n,p} \end{bmatrix} \qquad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix} \qquad \boldsymbol{\epsilon} = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}$$

In order to derive one of the methods to approximate the parameters $\boldsymbol{\beta}$, it is necessary to define **residuals**. A residual represents the deviation of the obtained approximation by the model from the true value of the target (Figure 1.1). When the model, defined by certain $\hat{\beta}$, approximates Y as $\hat{Y}$, then $Y - \hat{Y}$ is the residual.

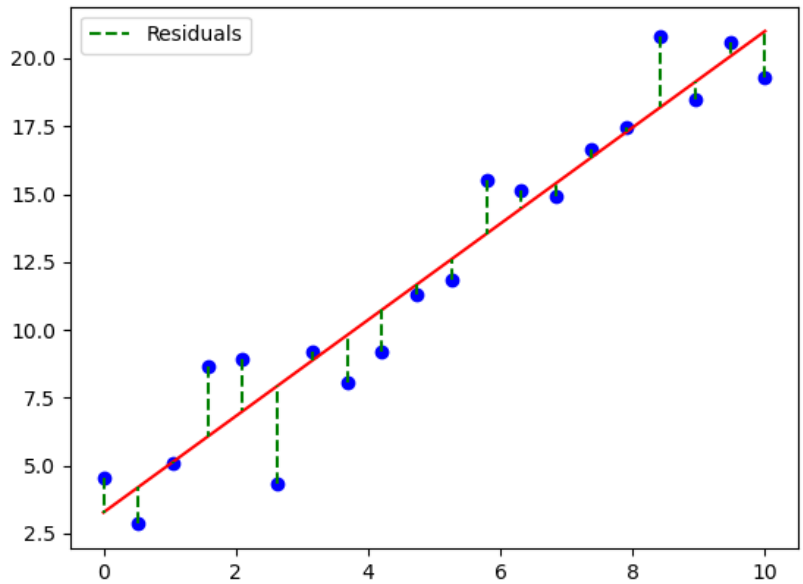

Figure 1.1: Illustrative example of residuals in the context of linear regression.

Using the residuals, we can measure the goodness of fit. We aim to define the linear relation to be as close as possible to the training data. In matrix notation, we can represent the residuals for all n available observations as $Y - \mathbf{X}\hat{\beta}$. To ensure that large errors are penalised more heavily and all of the residuals contribute positively, regardless of their sign, the general error function is defined as

$$(Y - \mathbf{X}\hat{\beta})^T(Y - \mathbf{X}\hat{\beta}). \tag{1.4}$$

Based on Equation 1.4, we can derive an optimization approach to derive $\hat{\beta} \approx \beta$ as

$$\min_{\hat{\beta}} (Y - \mathbf{X}\hat{\beta})^T(Y - \mathbf{X}\hat{\beta}). \tag{1.5}$$

The approach in Equation 1.5 stands for **least-squares estimation**. Other estimation methods for $\hat{\beta}$ exist; however, only the least-squares approach is included in the chapter, as it is used in the subsequent computational implementation.

1.3 Regularized variants of linear regression

Using solely linear regression might in a few cases result in overfitting.

1.4 k-Nearest neighbours algorithm

1.5 Random forest algorithm

1.6 Time-series analysis (ARIMA)

Chapter 2

Stock market data

The aim of this chapter is to introduce a set of related variables to the financial market, their possible importance in terms of predictive modelling, and lastly their subsequent processing into specific forms, which were used throughout all the related analyses in this thesis.

Market itself and particularly the price movement of financial instruments is a complex and unsurprisingly counterintuitive process with a large number of more or less related variables, which contribute to price movements. In terms of using such data as an auxiliary source for decision making in investment strategy creation, there is a rich universe of sources dealing with the issue [6], [11].

In the thesis, we aim mainly to research the predictive power of various more or less sophisticated methods of machine learning on the particular market segment. Many papers tend to focus on a single stock or a very limited set of a few stocks [7], [18]. Others tend to apply predictive modelling methods to stock funds *(e.g. US S&P 500 and NASDAQ or German DAX)* [15] however, there is a small number of sources dealing with a market segment, analyzing single stocks included in it.

Our main focus is on the index fund S&P 500 – a US stock index accumulating the 500 biggest publicly traded companies in terms of market capitalization. The fund itself is considered to be the most trustworthy general indicator of the overall US and world's stock economy state [9]. The fund itself is divided into several general market segments to which stocks belong. In this thesis, we focused on the segment **Information technology**. However, the approach used in our analyses is adaptable to any subset of stocks – the only determining constraint is the respective database of stocks.

The Information technology segment consists of 68 US-traded stocks. As of late 2025, in terms of total capitalization, this group makes a total of nearly 30 % of the fund, which makes it the most weighted sector, despite making only 13.6 % of all included stocks [13]. This recent growth of its weight could be explained by a lot of

factors steering the market nowadays, but the general view on the topic suggests that recent boom of artificial intelligence could be the most dominant engine of growth. This suggests that focusing on this segment is reasonable, as it could be considered the most important one nowadays.

2.1 Variables in financial data

At first, we introduce specific attributes which define the value movement of all financial and economic instruments used throughout the thesis as features and target variables. These attributes could be further generalized onto any financial instrument attribute's movement, not only to price itself – as it will be applied to non-stock data as well.

For the purposes of defining desired terminology, let's define a time interval on which we analyse an instrument's attribute movement as $[t_1, t_2]$, where naturally $t_1 < t_2$. Defined attributes, which are attached to a single time breakpoints, could be universally defined for any time interval with various frequency.

At the breakpoint t_1 we define the **opening value**, which denotes the value of the attribute at the very beginning of the interval. For the purposes of this thesis, where we aim to use a daily frequency of data, this value denotes the value of the attribute at the very start of the trading day. In the opposite manner, at the breakpoint t_2 we define the **closing value**, denoting the final value of the attribute at the end of the observed interval – in our case, at the end of the trading day.

During the process of value evolving throughout the interval, we observe the global minimum and maximum of the attribute's value, which we denote **low value** and **high value**, respectively. Such data give us an indication of volatility during the interval and subsequently the sentiment of the market regarding the attribute.

Last but not least, we accumulate data regarding the number of an instrument's stocks traded *(either sold or bought)* during the time interval. This is denoted as **volume**, and in terms of the thesis, this data is > 0 only for the stocks' data, as for some other instruments, this data is irrelevant and not tracked anyhow.

The last data attribute, applied only on stocks' data as well, which is strongly dependent on single stocks, is volume of **dividends** paid out. Some companies oblige themselves to pay parts of their profit to their investors on a regular basis *(e.g., Apple, Microsoft or Oracle)* [3]. With respect to the completeness of data fetched we included this data in our potential features database.

The visualisation of the terms for a sample interval of a Microsoft stock between the beginning of June 2025 and the beginning of September 2025 is pictured in Figure 2.1.

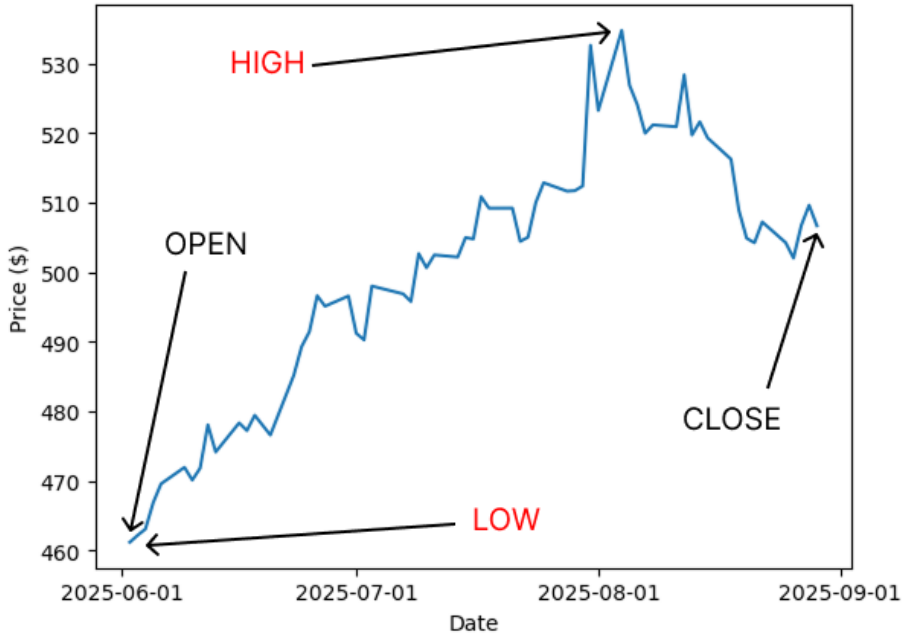

Figure 2.1: Visualisation of the defined attribute's movement terms on Microsoft stock price.

2.2 Rolling window and prediction methodologies

There are quite a few approaches on how to properly process time-series data into the respective training, validation, and test sets. In order to ensure next-day prediction, which means that by using trading days $t-(p-1)$, $t-(p-2)$, ..., t, we predict the price value of a financial instrument on the $t+1$ day, we can use the methodology of **rolling window** or commonly named as **sliding window** [4].

In this approach, we define a static time-series interval, which serves as a hybrid train-validation time interval. Let's define the overall data interval being between T_1 and T_2 breakpoints *(global interval)*. On this interval reside two frames of variables – multivariate $\mathbf{X} \in \mathbb{R}^{M \times F}$, where F denotes the number of features used, which serves as an input variable and univariate $Y \in \mathbb{R}^M$ accumulates values of the target variable to the respective rows of $\mathbf{X}$.

Further, we define parameter $1 \leq p < M$, which is the length of the training subwindow (look-back parameter) and w, which denotes the time units beyond which, we aim to determine predictions. For the purposes of this thesis and its next-day prediction methodology we define $w = 1$, but in the general auto-regressive case $w \geq 1$ is acceptable. With respect to the rolling window methodology, we can generalize the process of train and validation sets split, by creating a pseudocode demonstrating the relationship between input and target matrices. Algorithm 1 shows what parts of the

global $\mathbf{X}, Y$ are directly available at the time t.

Algorithm 1 Rolling window construction

We suppose $w \leftarrow 1$
for $t = T_1 + p$ to $T_2 - 1$ **do**
  $\mathbf{X}_t \leftarrow \mathbf{X}[t - p + 1 : t]$
  $Y_t \leftarrow Y[t - p + 1 : t]$
  Train respective model on $(\mathbf{X}_t, Y_t)$
  Generate one-day-ahead forecast $\hat{Y}_{t+1}$
end for

From Algorithm 1 a few important properties of this method could be derived. Firstly, respective intervals are overlapping and the shift between respective input and target matrices, at t and $t+1$, is a single time unit. Secondly, by ensuring overlapping, we maximize the number of possible training, thus validation, data points, making the training-validation process possibly more robust and general. Lastly, such setup would be implemented on train-validation interval (T_1, T_2) and on disjoint test interval likewise.

The multivariate $\mathbf{X}$ contains pre-chosen variables (Subchapter 2.3). In this thesis, time interval between 15th of September 2017 and 31st of December 2022 (1333 trading days) is used as training-validation interval and time interval between 1st of January 2023 and 16th of January 2026 (761 trading days) is used as test interval.

2.3 Suitable features in modeling

The amount of variables, which are directly or indirectly generated together with day-to-day market running, is quite enormous, and one should consider subsetting them for machine learning purposes wisely. In this subchapter, we aim to show and justify the process of our general feature selection, mostly based on a priori knowledge of processes within financial markets and the variables' most likely effect on stock markets.

For the sake of completeness, data were obtained from the API directly fetching data available on Yahoo Finance – available through the Python library abstraction provided by the `yfinance` library [1, 17].

2.3.1 Stock indices

In order to process the current general state of the stock exchange we researched possibilities of implementing data regarding the price development of certain index funds. **Index fund** is an abstract concept of accumulating information regarding

single stock assets into one unified asset, mostly using a predefined weighted average of prices of the stocks included.

The fund is specifically designed to follow and replicate the performance of a given subset of stocks. Taking this into account, by including such well-chosen funds, which copy a broader and more various scope of stocks on a given market, we may obtain valuable information regarding the current state of the US market.

For these purposes, we decided to include into our database two major stock indices. The general index **Standard's and Poor 500**, commonly known as S&P 500, is an index including the 500 biggest US-traded stocks based on total market capitalization. The index consists of a wide scope of various segments making it robust and universal in terms of mirroring the overall US market state.

Taking into account the scope of this thesis – the Information technology segment, we decided to include the index **NASDAQ Composite**, better known as NASDAQ. The main difference from S&P 500 is its segment focus. NASDAQ focuses mainly on tech and IT companies, making it possibly a valuable piece of data for our database of feature data.

2.3.2 Commodities

Commodities denote mainly raw resources and materials, which are traded mainly in uniform quality and large quantities. Their fundamental role in the market comes from the fact that most of these assets play an essential part in further processing, consumption and industry. Matsumoto et al. suggest, that the commodity market play a vital part in the whole financial segment, which suggests, that including such assets in the database might be sufficient [12].

For the purposes of this thesis, we included data regarding two major commodities – **crude oil** and **gold**. Even though there might not be a clear connection to the IT sector, we believe that such data would bring valuable information about the general state of the whole global economy.

Oil plays a vital role in keeping the world in motion, and a solid majority of the world's activities depend on its presence and availability. This might imply that these dependencies on the global market would be projected into the price movement of the commodity.

Last but not least, we introduce gold price movements into the database. For a wide range of reasons, gold still remains one of the most prominent securities. Most modern currencies are dependent on a system, where gold acts as a security stabilizing currency movements and limiting the governments issuing money. On top of that, governments rely on physical gold as a reserve in times of crises [14]. Taking these reasons into account, by including gold in the database of features, we assume we are

obtaining valuable information regarding the world's economy state, crisis potential or investors' sentiment, etc.

2.3.3 Currency pairs - FOREX

Currency pairs and subsequently the **Foreign Exchange Market** (FOREX) denote a specific market, where the main traded goods are currencies. Traders involved in the market speculate on certain exchange pairs' volatility and bet on them in order to make a profit. The market itself determines price of every possible currency pair, thus making the acquisition of certain amount of currency possible through various ways – currency X could be traded through currencies Y, Z, etc., making the market far more speculative than others. By the volume of instruments traded, this decentralized market is by far the largest market in the world [16].

Being fully decentralized, meaning prices on this market are accepted worldwide, these data could be considered as a sufficient indicator of currency relative strength and, by that, an indicator of a particular government's and economic competitivness of a certain country. Taking this into account, we assume, that including such data inside our database would bring valuable information about the international competitivness of US economy towards other major economies and quantitative information about relations among major economies, which is most likely an important predictor for stock price movements.

By our focus on the US market, we chose to include currency pairs with the American dollar (**USD**) in all pairs. We chose to include so-called **major currency pairs**, with respect to their overall traded volume on a daily basis, according to reports published by the **Bank for International Settlements** and to market conventions [5], [16].

In particular, included pairs, together with the USD, are the Euro **(EUR/USD)**, Japanese yen **(JPY/USD)**, British pound sterling **(GBP/USD)**, Swiss franc **(USD/CHF)**, Australian dollar **(AUD/USD)**, Canadian dollar **(USD/CAD)**, and New Zealand dollar **(NZD/USD)**.

2.3.4 Other market indicators

Apart from the introduced indicators and the general assets' price movement data above, there is still a vast group of possibly valuable features, which do not fit into the above-defined features' groups.

In order to track the performance of the US economy more precisely, it is wise to include information regarding national debt as possible features. For this purpose, we included data about **US Treasury bonds**. Bonds denote a financial instrument mostly long-term, low-risk securities issued by the government of the USA, where an

investor, in basic terms, lends money to the government in exchange for periodic fixed payments. These bonds are further divided based on their maturity – the time within which issuer obliges itself to fully repay issued money.

Price movements of bonds with various maturities are considered to be a good indicator of the US economy, and thus of the stock market state [10]. In our database of possible features we included bonds with a maturity of 13 weeks, which belong to the group of short terms bonds, and bonds with 5 year long maturity. This kind of maturity is on the edge between long-term and medium-term. The choice reflects the need to track data regarding relatively very short horizon and a long one.

Apart from the bond-related data we considered obtaining data which are not directly financial – they are not any type of asset. We researched possibilities of tracking the sentiment of investors and of the market itself. We chose to include data of **Volatility Index** *(VIX)*, which is a specific gauge tracking state of the US stock market in terms of volatility and the sentiment of investors.

It presumably brings valuable information, which might not be included in all of the metrics described, regarding the a priori mood of the market and its assets' risk. Its value ranges mostly up to 80, which was considered as crisis values in years 2020 and 2008, and above approximately 10, which is considered to be relatively stable market state. Value development between the January and September of 2025 is given in Figure 2.2.

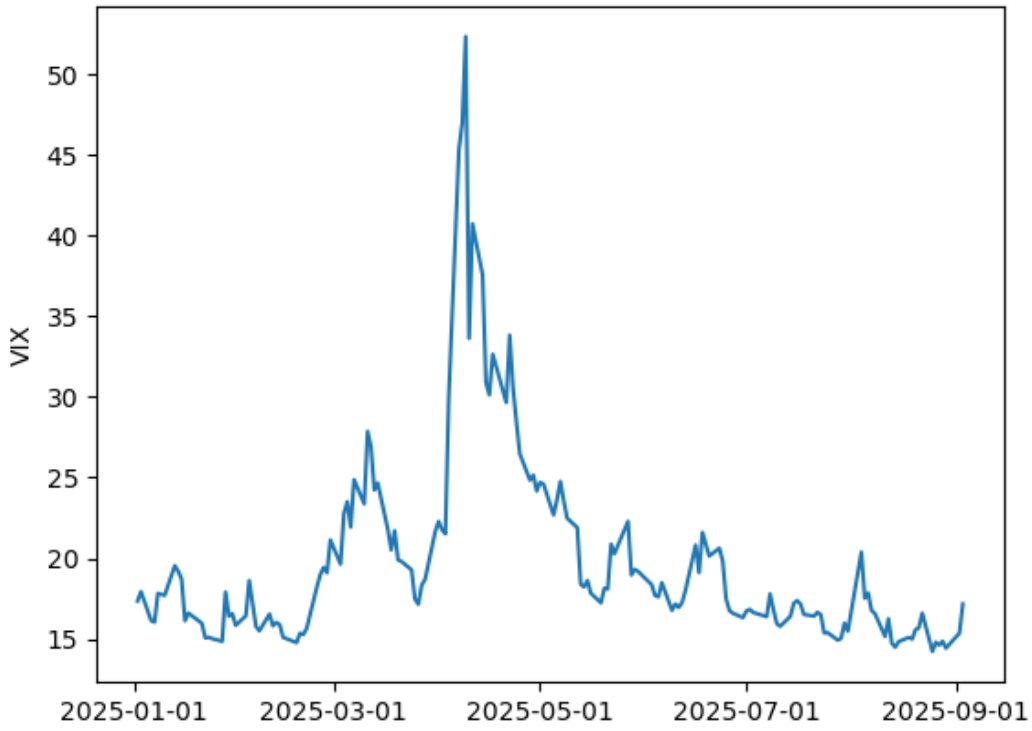

Figure 2.2: VIX value movement through the 2025.

It is surely interesting to address the sudden VIX surge during the April 2025. This sudden increase was basically day-to-day and denotes phase of repeated issuing and cancelling of international tarrifs. As we can see, this political move damaged investors' exposure to the US market then.

Chapter 3

Predictive analyses on IT segment

This chapter will be dedicated to certain results obtained during training and validation of several algorithms and approaches on the data. We plan to analyze results brought by included algorithms in first chapter in detailed manner.

Chapter 4

Intraday optimization of portfolio based on model's prediction

This chapter will deal with the topic of using the obtained data predictions *(preferably from the most precise model)* to optimize weights of single stock assets in hypothetical investment portfolio. For these purposes, we introduce Markowitz portfolio theory and the optimization problem coming out of it. With such foundations we analyze results in terms of profit and risk by our models.

Conclusion

This section will be dedicated to concluding our results, accumulating them into a concise form, evaluating successfulness of completing given goals and suggesting future tweaks in order to make thesis' idea more robust and practice-friendly.

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Appendix A: Source code of the model

Appendix attached to this thesis contains all the source codes used in analyses introduces in the thesis. All the source codes and results are freely available as a Git repository at `https://github.com/sugy-w/Bachelor-Thesis`.

Please note, that all the contents of this repository is being shared only for academic purposes and its modifying, redistribution or commercial use is strictly prohibited by the included licenses.